

SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – 26060
- **Problem Statement Title-** Digital Twin Platform for India's Antarctic Research Stations (Maitri & Bharati)
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)** Commit Crew



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

A live digital twin of each station (buildings, generators, fuel, stores, weather and crew) shared by the NCPOR control room in Goa and the crew on the ice.

❖ Detailed explanation of the proposed solution

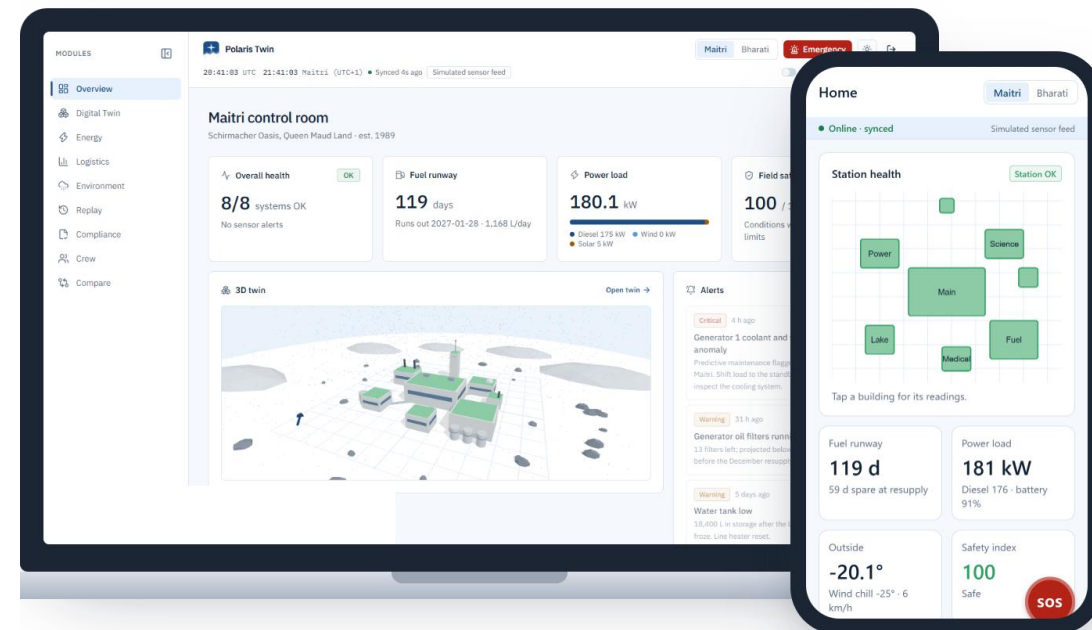
- **Web dashboard (NCPOR Goa):** 3D twin, energy, logistics, environment, replay, compliance, compare, emergency mode.
- **Crew Android app:** alerts, 2-second-hold SOS, field check-out, inventory scanning, wellbeing check-in.
- **Ask Polaris copilot:** Gemini answers in English / हिंदी from live data and station SOPs.

❖ How it addresses the problem

- Integrates infrastructure, energy, logistics and environment in one model for remote management.
- Predicts fuel runway and generator faults before they become emergencies.
- Edge-first: the station keeps working when the satellite link drops; ≈ 0.5 KB delta syncs.

❖ Innovation and uniqueness

- Offline queue with SOS priority, built for Antarctic connectivity.
- One shared logic core: web, app and copilot always show the same numbers.
- Ship-delay and cold-snap what-ifs, explainable alerts with time-to-limit.



Working prototype, live today

polaris-twin-psi.vercel.app

Web dashboard + Android APK (Expo)

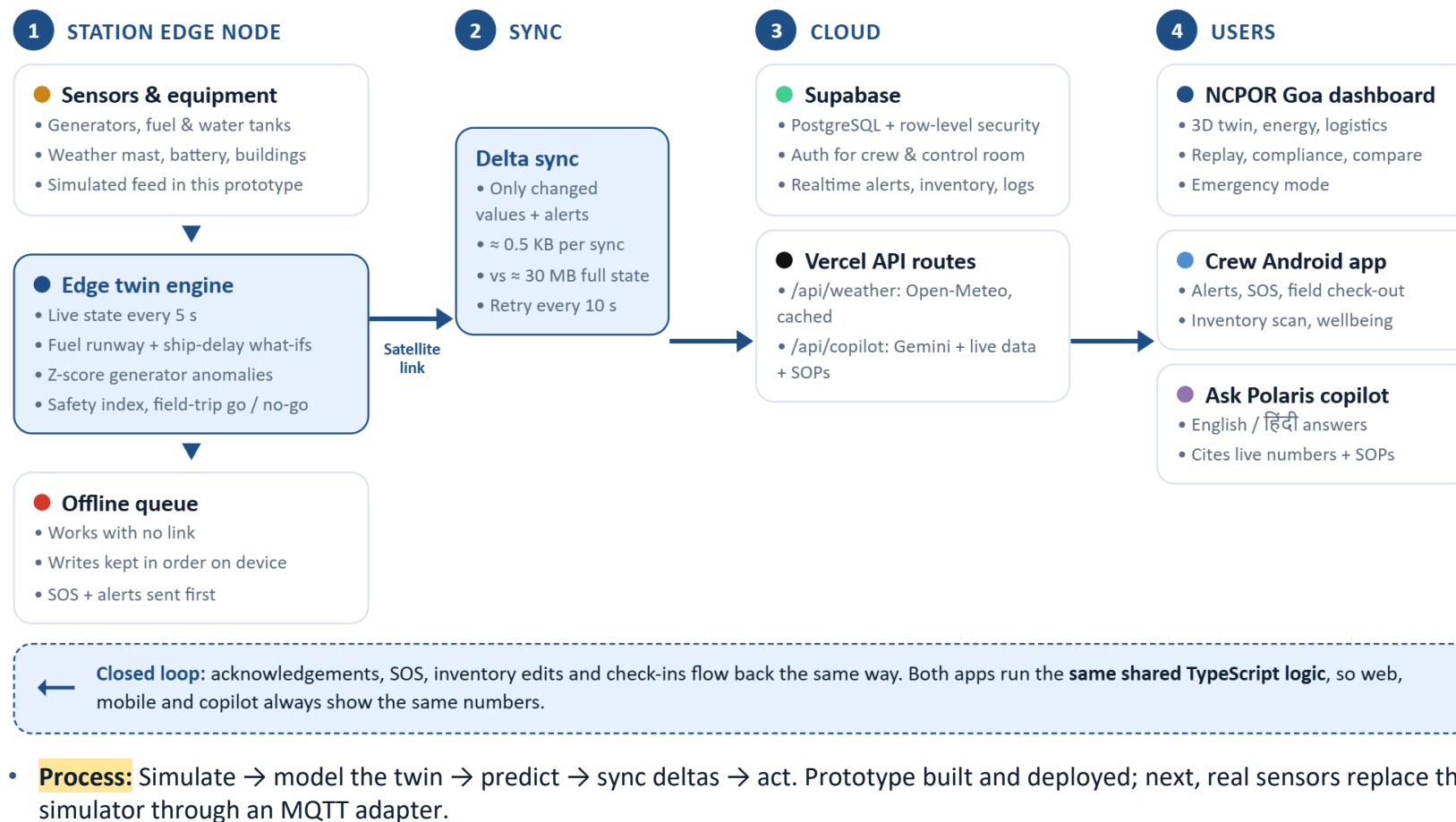
Simulated sensor feed; real Open-Meteo weather



❖ Technologies to be used

-  **Next.js 16**
Web dashboard + API
-  **React · R3F / three.js**
3D digital twin
-  **TypeScript**
Shared logic core
-  **Tailwind CSS · shadcn/ui**
UI · Recharts charts
-  **Expo (React Native)**
Crew app · EAS build
-  **Supabase**
Auth · Realtime
-  **PostgreSQL**
Row-level security
-  **Google Gemini**
Ask Polaris copilot
-  **Vercel**
Hosting · Open-Meteo proxy

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



❖ Analysis of the feasibility of the idea

- **Technical:** end-to-end prototype already running: live website, Android APK, realtime database and AI copilot.
- **Bandwidth:** ≈ 0.5 KB delta per sync instead of ≈ 30 MB, fits a constrained satellite link.
- **Cost:** open-source frameworks and free tiers; one small edge box per station.
- **Operational:** dashboard and app need no special hardware; works in English and Hindi.

❖ Phased rollout

- Phase 1: pilot at Maitri with generator, fuel and weather sensors.
- Phase 2: add Bharati and inventory linked to the ISEA ship manifest.
- Phase 3: full building systems, maintenance history, offline voice copilot.

❖ Potential challenges and risks

Satellite link drops

Hours of outage; low bandwidth

Legacy equipment

Generators & tanks speak different protocols

False alarms

Noisy sensors erode trust

Extreme cold & power

-35°C , diesel-limited power

Data security

Critical national infrastructure

❖ Strategies for overcoming these challenges

Edge-first + offline queue

Station keeps working; deltas sync later

MQTT adapter layer

One schema; simulator swaps out per sensor

Explainable z-scores + human ack

Each alert says why and what to do

Low-power rugged edge box

Runs on station UPS; tiny payloads

Auth + row-level security

Keys never in code; audit-ready logs

2 apps

web + Android, live

9

dashboard modules
built

≈ 0.5 KB

per satellite sync

❖ Potential impact on the target audience

- **NCPOR Goa:** live view of both stations, early warnings, one-click compliance reports.
- **Station leader & crew:** SOS, overdue-trip alarms, go / no-go before every field trip.
- **Logistics planners (ISEA):** know what runs out before resupply and plan the ship load.
- **MoES:** data-backed planning for future stations and budgets.

❖ Benefits of the solution (social, economic, environmental, etc.)

- **Social:** safer field work, faster emergency response, anonymous crew wellbeing tracking.
- **Economic:** less diesel waste, fewer breakdowns, fewer emergency resupplies.
- **Environmental:** CO₂, waste and spill tracking under the Madrid Protocol and Indian Antarctic Act 2022.
- **Strategic:** strengthens India's polar research presence with home-grown technology.

Today vs Polaris Twin

	TODAY	WITH POLARIS TWIN
 Station status in Goa	Periodic manual reports	Live, refreshed every 5 s
 Fuel shortfall	Seen when stocks run low	Runway + delay what-ifs, months ahead
 Generator faults	Found after breakdown	Anomaly alert with time-to-limit
 Field trips	Judgement + radio	Go / no-go from 48 h forecast
 Sync over satellite	Full files, ≈ 30 MB	Deltas, ≈ 0.5 KB
 Compliance	Paper registers	CO₂, waste, spills, one-click report

5 s

live refresh in Goa

60,000 ×

smaller syncs than full state

48 h

forecast behind go / no-go

❖ Details / Links of the reference and research work

Domain & policy

- NCPOR, Ministry of Earth Sciences: ncpor.res.in
- Maitri station: [en.wikipedia.org/wiki/Maitri_\(research_station\)](https://en.wikipedia.org/wiki/Maitri_(research_station))
- Bharati station: [en.wikipedia.org/wiki/Bharati_\(research_station\)](https://en.wikipedia.org/wiki/Bharati_(research_station))
- Protocol on Environmental Protection (Madrid Protocol): ats.aq/e/protocol.html
- Indian Antarctic Act, 2022, Ministry of Earth Sciences: moes.gov.in
- COMNAP, Council of Managers of National Antarctic Programs: comnap.aq

Research

- M. Grieves, Digital Twin: Manufacturing Excellence through Virtual Factory Replication (white paper, 2014)
- Digital twin overview: en.wikipedia.org/wiki/Digital_twin

Technology documentation

- Open-Meteo weather API: open-meteo.com/en/docs
- Supabase Realtime: supabase.com/docs/guides/realtime
- Next.js: nextjs.org/docs
- Expo EAS Build: docs.expo.dev/build/introduction
- Gemini API: ai.google.dev/gemini-api/docs

Our prototype

- Live dashboard: polaris-twin-psi.vercel.app
- Demo login shown on the sign-in page



Scan for the live prototype